

CE8014 – Agentic AI: Foundations and Development

Assignment 3: Autonomous Multi-Doc Financial Analyst

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1 Embedding Model Comparison

1.1 Setup

Three embedding models were leveraged to embed the text corpus for vector database building and text input questions for retrieval, including:

- **minilm**: `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` – 118 M parameters, multilingual, optimized for paraphrase similarity.
- **mpnet**: `sentence-transformers/all-mpnet-base-v2` – 109 M parameters, English-focused, trained on 1 B+ sentence pairs.
- **gemma_embed**: `google/embeddinggemma-300m` – 300 M parameters, Google’s instruction-tuned embedding model, multilingual.

The default setting for these embeddings are GRAPH mode, `chunk_size` = 2000, and `chunk_overlap` = 500. The LLM API is `moonshotai/kimi-k2-instruct-0905` from Groq.

1.2 Results

Model	Parameters	Multilingual	Score	Accuracy
minilm	118 M	✓	9 / 14	64%
mpnet	109 M	×	10 / 14	71.4%
gemma_embed	300 M	✓	11 / 14	78.6%

Table 1: Embedding model comparison at `chunk_size` = 2000.

gemma_embed outperforms the others. The accuracy of the pipeline using `gemma_embed` is slightly better than the others. This can be explained due to its bigger scale; `gemma_embed` can bring a better text representation, resulting in more accurate retrieval.

minilm is fast but less precise. Despite being multilingual, `minilm` consistently underperformed at 64%. The paraphrase-similarity objective of MiniLM is well-suited for semantic matching of short sentences but less effective for financial table retrieval, where the query (e.g., “Capital Expenditures 2024”) must align with dense numeric tables rather than prose.

mpnet is competitive for English queries. All failing test cases for `mpnet` are the same four cases (Tests B, E, A2, B2) that fail across other models. Thus, it is clear that `mpnet` has quite good capacity despite its lack of support for multilingual.

2 LangGraph vs. LangChain (ReAct)

2.1 Setup

Due to better performance compared to other embeddings, both agents were evaluated on the identical 14-question benchmark suite using `gemma_embed` embeddings and the default setting mentioned above. There are two modes for LangGraph-based and LangChain-based agents, respectively:

- **GRAPH mode:** LangGraph state machine (Tasks B–E) with a retrieve → grade → [rewrite loop / generate] flow.
- **LEGACY mode:** LangChain `create_react_agent` with `AgentExecutor` and `max_iterations = 5`.

2.2 Results

Agent	Score	Accuracy	Trap Tests	Avg. Latency
LangGraph (GRAPH)	11 / 14	78.6%	PASS (3/3)	~10–35 s/query
LangChain (LEGACY)	10 / 14	71.4%	PASS (3/3)	~2–60 s/query

Table 2: LangGraph vs. LangChain comparison (`gemma_embed`, `chunk.size = 2000`).

The four failing tests (B, E, A2, B2) are identical for both agents, confirming that these failures originate from incorrect or ambiguous ground-truth expected values in the test suite, not from the agent architecture.

Trace Example: Self-Correction in Action. The following trace (from Test B, GRAPH mode) demonstrates the grader triggering a rewrite cycle. The original Chinese query about Tesla R&D returned cost-of-revenue chunks (irrelevant), so the grader outputs `no`. Then, the rewriter produces a more precise English query, and the second retrieval successfully fetches the R&D expense table.

```
> Test B: Tesla R&D (Chinese)
--- RETRIEVE ---
Routed to: tesla
[Reranker] scores: [1.006, 0.39, -3.783, -5.219, -5.38]
--- GRADE ---
Grade: no
--- REWRITE ---
Rewritten: Tesla research and development expenses fiscal year 2024
--- RETRIEVE ---
Routed to: tesla
[Reranker] scores: [4.099, 2.528, 0.059, -4.204, -6.996]
chunk[0] page=42 | R&D $ 4,540 $ 3,969 $ 3,075
--- GRADE ---
Grade: yes
--- GENERATE ---
A: Tesla's R&D expense for FY2024 was $4,540 million.
```

Listing 1: LangGraph self-correction trace for Test B.

The LangChain LEGACY agent handles the same test in a single tool call without any rewrite cycle, relying on the ReAct loop’s internal reasoning to identify the correct passage. Both succeed, but the LangGraph path is more transparent and auditable.

2.3 Architectural Comparison

Property	LangGraph (GRAPH)	LangChain ReAct (LEGACY)
Control flow	Explicit state machine; each node has a single responsibility	Single LLM loop; agent decides all transitions implicitly
Self-correction	Structured: grader outputs yes/no , rewriter rewrites, loop retries	Implicit: agent may retry or give up based on scratchpad
State management	<code>TypedDict</code> with typed fields (<code>question</code> , <code>documents</code> , <code>search_count</code> , <code>needs_rewrite</code>)	Unstructured scratchpad accumulating Thought/Action/Observation text
Retry logic	Deterministic: max 2 rewrites, then forced generation	Non-deterministic: up to <code>max_iterations=5</code> , agent may stop earlier
Debuggability	Each node is a standalone function, individually testable	Parsing errors in the scratchpad can cascade silently
Latency	Predictable per-node cost	Variable; scales with number of tool calls and scratchpad length
Multilingual	Router handles Chinese queries explicitly	Relies on the ReAct prompt’s instruction to answer in English

Table 3: Qualitative comparison of the two agent architectures.

3 Chunk Size Trade-off

3.1 Setup

All settings mentioned in Section 2 except `chunk_size`, and `chunk_overlap` are reused. Pairs of (`chunk_size`, and `chunk_overlap`) specified for these experiments are (1000, 200), (1600, 400), and (2000, 500).

3.2 Results

Chunk Size	Overlap	Score	Accuracy	Observation
1000	200	10 / 14	71.4%	Faster retrieval, some tables split
1600	400	10 / 14	71.4%	Intermediate; no change
2000	500	11 / 14	78.6%	Best; full tables fit in one chunk

Table 4: Chunk size comparison using `gemma_embed`, GRAPH mode.

3.3 Context Precision vs. Context Completeness

Context Precision (small chunks, e.g., 500–1000 chars). Each retrieved chunk is focused and contains a few irrelevant numbers. The LLM is less likely to pick the wrong figure from a dense passage. A small chunk will fracture these tables mid-row, causing the LLM to receive a partial view with missing column headers or cutoff values.

Context Completeness (large chunks, e.g., 2000 chars). An entire income statement table fits within one chunk, giving the LLM visibility into all years’ columns simultaneously. This is essential for cross-year disambiguation (e.g., “2024 vs. 2023 Automotive Sales”). The downside is that the chunk may include multiple unrelated line items, increasing the risk that the LLM extracts the wrong figure. With a strong embedding model (`gemma_embed`) and the generator’s explicit year-alignment rule (Task E, Rule 5), this risk is mitigated.

References

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